EDWIN SALVADOR SOTO HERNANDEZ

Software Developer — Networking & IoT Engineer

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Professional Summary

Software developer with a background in Information and Communications Technology and an interest in software engineering, network infrastructure, connected systems, and data analysis. Experienced in applying technical knowledge to academic and independent projects, with a focus on problem-solving, collaboration, and continuous improvement.

Experience

Network Infrastructure & Security Practitioner 2024 – Present
ITSOEH – Hidalgo, Mexico
Cisco Networking Laboratory (Academic Experience)
- Configure and troubleshoot IPv4/IPv6 networks, implementing VLANs, 802.1Q trunking, inter-VLAN routing, EtherChannel, STP, DHCP, and dynamic routing protocols across multi-router topologies.
- Implement network services and enterprise security measures, including DHCPv6, HSRPv2, ACLs, port security, SSH, and device hardening, while exploring network automation and security.

Technologies: Cisco IOS, IPv4/IPv6, VLANs, STP, EtherChannel, DHCP/DHCPv6, HSRP, ACLs, SSH, Cisco Packet Tracer.

IoT Workshop Instructor & Competition Preparation Lead 2024 - Present
ITSOEH IoT Club – Technical Workshops & State-Level Competitions
- Led hands-on technology workshops for students, introducing programming, IoT and project development.
- Trained and mentored student teams preparing for state-level technology and robotics competitions.
- Supported the development of technical skills through collaborative projects and competition preparation.

Technologies: MicroPython, Raspberry Pi, Microcontrollers, IoT Sensors, LoRa, MQTT, Linux, Node.js, WebSockets.

Data Analytics & Predictive Modeling Collaborator March 2025
XVI National Robotics and STEM Skills Championship – Hidalgo, Mexico
- Collaborated on the "Aquaculture Pond Monitoring System" project, contributing to data analysis and predictive modeling to optimize sustainable fish production.
- Applied Time-Series forecasting models using Prophet to analyze sensor data and predict aquaculture conditions.

Technologies: Python, Prophet, IoT Sensor Data, Time-Series Forecasting, Data Analysis

Projects

RehabiNodes: Wireless Rehabilitation Monitoring System • InnovaTecNM 2026 Prototype
Wireless sensor-based platform for real-time monitoring of arm/forearm movement and muscle activity, featuring 3D visualization and historical data storage to support physical rehabilitation.
ESP32 · IMU Sensors · LoRa · MQTT · Docker · Node-RED · InfluxDB · Grafana · Three.js

IoT Aquaculture Pond Monitoring System • XVI National Robotics and STEM Skills Championship 1st Place
IoT-based system for monitoring aquaculture pond parameters via connected sensors, transmitting data to a digital platform for sustainable production analysis; awarded 1st Place in the Innovative IoT Project category.
IoT Sensors · Wireless Communication · Python · Prophet · Data Analysis · Time-Series Forecasting

Converged Network Infrastructure Implementation • ITSOEH Academic Practicum Staging Environment
Designed and configured converged network topologies with VLAN segmentation, inter-VLAN routing, and link aggregation (EtherChannel) to ensure high availability and security in an enterprise lab environment.
Cisco IOS · IPv6 · DHCPv6 · HSRPv2 · OSPF/Static Routing · VLANs · SSH

CCRT Registration System • Centro Cultural Regional Tlahuelilpan (CCRT) Production-Ready
Web application for comprehensive management of students, teachers, and workshops, including record tracking, progress monitoring, and automated PDF generation for graduates.
C# · ASP.NET Core 8 MVC · Entity Framework Core · SQL Server

T-Box Olinia: Vehicle Telemetry Unit • Retos Olinia – InnovaTecNM 2026 Prototype
Designed hardware and firmware for a low-cost vehicle telemetry unit (T-Box) based on ESP32-C6, featuring CAN bus (ISO 11898) data acquisition, 4G connectivity (MQTT), and native Zigbee mesh networking. Developed hybrid communication architecture, PCB layout, and IP54-rated enclosure.
ESP32-C6 · C/C++ · CAN Bus · MQTT · Zigbee · PCB Design

LogScore • Crevolution Hackathon 2026 2nd Place Prototype
Financial inclusion and alternative credit scoring platform. Interactive frontend prototype for habit management and 'behavioral collateral', using local OCR for ticket digitization and risk scoring estimation.
HTML5 · CSS3 · JavaScript (ES6+) · Fintech · OCR · UX/UI

Education and Training

B.Eng. in Information and Communications Technology 2023 - 2027 (Expected)
Instituto Tecnológico Superior del Occidente del Estado de Hidalgo (ITSOEH) — Hidalgo, Mexico

Samsung Innovation Campus 2026 2026 – 2027 (Expected)
Universidad de Monterrey (UDEM) — Mexico

Master in Data Science with AI Feb – May 2026
Centro Público de Formación en Inteligencia Artificial (INFOTEC) — Mexico

Certifications:
GitHub Actions Certification (GH-200)

Additional Certificates:
CCNA (Cisco), Data analytics (Google), IoT (Cisco & Huawei), Linux Server Administration (Cisco), HCIA (Huawei)

Languages
Spanish: Professional proficiency (Native)
English: Intermediate professional proficiency

Technical Skills

Programming & Frameworks: JavaScript (ES6+), TypeScript, Python, C#, Java, C/C++, React, React Native, Node.js, ASP.NET Core MVC, Entity Framework Core
IoT & Embedded: ESP32 / ESP32-C6, Raspberry Pi, MicroPython, GPIO, IMU/EMG Sensors, LoRa, Zigbee, CAN Bus (ISO 11898), MQTT
Networking: Cisco IOS, IPv4/IPv6, VLANs, STP, EtherChannel, DHCP/DHCPv6, HSRP, OSPF, ACLs, SSH, Port Security
Cloud, DevOps & Tools: Docker, Linux, Git, GitHub, GitHub Actions, Virtual Machines, REST APIs, Cloud Fundamentals
Databases: SQL Server, MySQL, MariaDB, MongoDB, InfluxDB
Data & Analytics: Pandas, Prophet, Time-Series Forecasting, Linear Regression
Architecture & Design: MVC, Microservices, SOLID, DRY, KISS, GoF Patterns (Builder, Decorator, Observer)
Methodologies: Agile, Scrum, 5E Model
Soft Skills: Teamwork, Leadership, Communication, Critical Thinking, Time Management, Adaptability, Problem-Solving, Collaboration